

VANUA BALAVU, Fiji

WMO: 916760

Lat: 17.233S

Lon: 178.950W

Elev: 52

StdP: 100.70

Time Zone: 12.00 (PCF)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.4	21.9	13.8	9.9	22.9	14.9	10.6	23.2	11.6	22.9	11.1	23.1	5.7	130	0.391

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	3.9	31.0	26.8	30.4	26.6	30.0	26.3	28.7	29.5	27.9	29.2	27.4	28.8	4.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.5	25.0	29.1	27.6	23.7	28.5	27.0	22.8	28.1	93.7	29.5	89.8	29.3	87.2	28.7	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	9.5	8.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.1	27.3	27.6	27.6	26.9	25.8	25.0	24.5	24.6	24.8	25.4	26.3	26.9		
	DBStd	1.58	1.10	1.05	0.93	1.06	1.05	0.95	1.09	1.27	1.24	1.19	0.95	1.19		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD10.0	5860	537	493	545	507	489	451	448	453	444	479	488	524		
	CDD18.3	2818	279	259	287	257	231	201	190	195	194	221	238	266		
	CDH23.3	22591	2842	2746	3002	2441	1703	1128	824	917	1019	1442	1988	2539		
(13)	CDH26.7	3719	666	713	739	434	141	37	23	50	47	126	237	507		
(14)	Wind	WSAvg	4.0	3.1	2.8	3.2	3.9	4.0	5.0	5.2	5.0	4.7	3.9	3.6		
(15) Precipitation	PrecAvg	1812	237	192	214	190	142	110	72	74	95	133	145	209		
	PrecMax	2424	486	483	499	427	265	320	126	176	234	298	332	404		
	PrecMin	1240	45	21	24	15	37	14	8	18	23	29	40	27		
	PrecStd	321	116	111	108	99	69	73	33	42	55	64	75	91		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.5	31.7	31.6	30.9	29.6	28.6	28.1	29.2	28.7	30.0	30.1	31.4		
		MCWB	27.1	27.6	27.3	27.1	26.2	25.1	23.9	25.6	24.8	26.3	25.9	26.9		
	2%	DB	30.6	30.7	30.6	30.0	28.7	27.6	27.3	28.0	27.7	28.8	29.2	30.4		
		MCWB	26.5	26.8	26.8	26.6	25.0	24.5	23.8	24.9	24.1	25.1	25.5	26.5		
	5%	DB	30.0	30.2	30.0	29.4	28.1	27.0	26.7	27.1	27.1	27.9	28.6	29.6		
		MCWB	26.3	26.4	26.4	26.0	24.4	23.9	23.6	24.1	23.8	24.4	25.1	26.1		
	10%	DB	29.4	29.6	29.5	28.8	27.5	26.5	26.2	26.4	26.6	27.2	28.0	28.9		
		MCWB	26.0	26.1	26.1	25.5	24.1	23.6	23.3	23.3	23.5	23.9	24.7	25.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.4	29.3	29.3	29.0	27.1	26.1	26.5	26.4	26.2	26.5	28.0	28.9		
		MCDB	29.8	30.1	29.9	29.5	28.2	27.5	26.7	28.3	27.7	29.1	28.7	29.4		
	2%	WB	28.3	28.2	28.1	27.8	26.0	25.4	25.4	25.5	25.3	25.7	27.0	27.7		
		MCDB	29.2	29.6	29.5	28.8	27.8	26.8	25.8	27.3	26.8	27.8	28.0	29.0		
	5%	WB	27.3	27.5	27.3	27.1	25.4	24.9	24.7	24.7	24.6	25.2	26.2	27.1		
		MCDB	28.9	29.1	29.0	28.2	27.2	26.3	25.5	26.5	26.3	27.1	27.3	28.4		
	10%	WB	26.6	26.8	26.7	26.5	24.8	24.5	24.0	24.1	24.1	24.8	25.7	26.4		
		MCDB	28.6	28.6	28.6	27.9	26.8	25.9	25.6	25.8	25.8	26.6	27.0	28.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	3.8	3.9	3.8	3.4	3.1	2.7	2.9	2.9	3.0	3.2	3.3	3.5		
		MCDBR	4.4	4.4	4.3	3.9	3.8	3.2	3.4	3.6	3.6	3.8	3.8	4.2		
		MCWBR	2.3	2.4	2.3	2.3	2.0	1.9	2.0	1.9	2.0	2.1	2.1	2.3		
	5% WB	MCDBR	4.0	4.1	4.2	3.9	3.5	3.1	3.0	3.5	3.3	3.5	3.5	3.8		
		MCWBR	2.7	2.7	2.7	2.7	2.1	1.9	2.0	1.9	2.1	2.1	2.3	2.7		
(40) Clear-Sky Solar Irradiance	taub		0.385	0.380	0.377	0.388	0.386	0.387	0.379	0.385	0.400	0.388	0.385	0.388		
	taud		2.630	2.655	2.666	2.624	2.608	2.589	2.598	2.580	2.538	2.576	2.607	2.606		
	Ebn at Noon		960	955	936	889	855	835	854	879	899	935	953	956		
	Edh at Noon		102	98	94	92	88	87	88	95	105	105	103	104		
(44) All-Sky Solar Radiation	RadAvg		6.30	6.05	5.66	4.84	4.28	3.84	4.13	4.67	5.33	5.87	6.09	6.12		
	RadStd		0.59	0.53	0.41	0.39	0.30	0.31	0.21	0.41	0.37	0.44	0.52	0.43		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page